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Enhancing green energy security: exploring multi-agent systems perspectives for a secure smart grid information System

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Abstract

In a world increasingly dependent on green energy and interconnected systems, the energy security has become a top priority. Smart grid technology is one of the most important green innovations in this area. It promises efficient energy management and an increased use of renewable resources. However, given the various vulnerabilities in smart grids and the history of cyber-attacks on energy systems, securing their communications is becoming more important than ever. This article examines the importance of securing smart grid information systems. It also explores the similarities in structure and function between multi-agent systems and an interconnected Microgrids network in a smart grid environment. In addition, different types of cyber threats to smart grid information systems are discussed and various existing approaches for detecting and mitigating these attacks are outlined. The paper concludes by suggesting that the use of Multi-Agent Systems principles in smart grid information systems can lead to innovative solutions for addressing their collaboration security challenges in the face of cyber threats, thus contributing to a greener and more resilient future.

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1. Introduction

As the world evolves towards an increasingly connected society with a high dependence on green energy, the quest for resilient and sustainable energy security has become more important than ever. A key indicator of this evolution is

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the transition from a traditional power grid to an interconnected intelligent infrastructure (Smart grids) based on distributed energy resources (DERs) and Microgrids. This evolution promises not only efficient management of electrical energy (production and consumption), but also greater use of renewable energy resources. The smart grid is a green technology that offers flexibility and scalability in the field of sustainable energy management by enabling its components to collaborate efficiently, and optimize their exchanges. However, this extensive interconnection raises cybersecurity concerns due to the multiplication of potential entry points for cyber-attacks.

To meet these challenges, researchers and practitioners have explored various solutions. One of the most promising ideas is the implementation of Multi-Agent System (MAS) collaboration models and mechanisms to secure smart grid information systems.

In fact, the researches in recent years have been marked by significant advances in the fields of smart grids and multi-agent systems (MAS) [1–3], representing a dynamic state of the art in energy distribution and smart systems management. Smart grids, characterized by interconnected Microgrids [4], have evolved to meet the growing demand for resilient and sustainable energy solutions. At the same time, MASs have seen significant progress [5–7], with advanced algorithms and autonomous decision-making capabilities becoming solutions to complex problems. Notable developments, such as reinforcement learning in MAS and decentralized decision-making protocols, have contributed to their adaptability and collaborative potential. The intersection of smart grids and multi-agent systems is fertile ground for research [8, 1, 3], with ongoing investigations into collaborative energy management, cybersecurity challenges [9–11], and distributed energy resource optimization [12]. Due to page restrictions, a more detailed state-of-art will be the subject of a future paper that is currently in progress.

In order to contribute to the field of securing green technologies, the goal of this paper is first to show the relevance of securing smart grid information systems, then to demonstrate the similarities that exist between multi-agent systems and smart grids, before exploring the opportunities offered by this analogy.

Our methodology is then based on qualitative research design, to investigate the role of multi-agent systems (MAS) in enhancing smart grid information systems and improving green energy security. The methodology includes a thorough review of researches in the smart grid security and the potential applications of multi-agent systems. Through thematic analysis, we identify emerging patterns and themes in the qualitative data collected from the literature review, enabling the construction of a sound theoretical framework for our future research in the field. In line with ethical guidelines, this research has attempted to triangulate data from multiple sources as best as possible, to ensure credibility and reliability.

The rest of this paper is organized as follows: Section 2 provides a background on the smart grids and microgrids applications as main green technologies. The security of sustainable energy is discussed in Section 3. Then, a general overview of cyberthreats to smart grids, is given in Section 4, followed by an analysis of specific cyberattacks on smart grid information systems in Section 5. Section 6 discusses different approaches for detecting and mitigating cyberattacks on smart grids. Section 7 presents the similarity between multi-agent systems (MAS) and smart grids (SG), followed by some conclusions. Finally, we conclude our paper with remarks and main recommendations, for future work directions.

2. Green technologies and applications (Smart grids and Microgrids)

Achieving sustainable and resilient energy security requires a holistic approach, involving cooperation between various political and economic stakeholders, as well as the development of green, secure and appropriate technological solutions.

From a technological point of view, there are a number of innovative and environmentally friendly solutions in the field of sustainable energy security, of which Smart Grids and Microgrids are ones of the most encouraging.

Smart grid and microgrid technologies each have their own respective applications. However, it is important to understand that both grid types are essential components of grid resiliency and reliability.

2.1. Smart grids: Characteristics and advantages

Smart grids are defined as networks characterized by a bidirectional flow of electrical energy and information, which depend on information and communication technologies ICT, in order to establish a smart decentralized electric power network [1].

In fact, the smart grid uses two-way communication technologies, control systems and computer processing to improve grid reliability. The concept also includes advanced sensors that display the grid stability, smart meters, and different types of power generation systems. Other capabilities of the smart grid include reporting failures, using relays to recover automatically from faults, and implementing automated feeder switches that redirect power around grid issues [13].

The advantages of implementing smart grid notions with power generation systems like Microgrids are numerous, including; more efficient electricity transmission; quicker and more effective responses to power disruptions, improved grid balancing and management (particularly during peak demand periods), a greater capability to integrate more renewable and DERs, and better security. Additionally, it will increase consumer involvement, improve reliability, lower costs, and contribute to a safer, green environment. For this reason, smart grids are essential to the success of the "Green Smart Cities" concept, enabling cities to reduce their carbon footprint.

Finally, Smart grids contribute significantly to green energy security by strengthening resilience, promoting the use of renewable energy, optimizing demand management and providing greater flexibility in the face of uncertain supply conditions.

2.2. Microgrids

Microgrids are autonomous and self-controlled energy systems (generation and load), integrating different types of renewable energy sources, capable of operating either in collaborative or isolated mode, making them more resilient in the face of outages and natural disasters (such as Hurricane Sandy in 2012 in the USA which drove massive interest in microgrids) [12].

A microgrid enables the simultaneous integration of interconnected loads and distributed energy resources that operate in parallel with the primary power grid, or in autonomous "islanded" mode [14]. This feature enables the microgrid to be isolated from the main distribution system in the event of a power failure, without compromising the integrity of the transmission infrastructure. In addition, it facilitates the rapid integration of other Distributed Energy Resources (DER), without the need to rethink the control architecture of the distribution system [4].

In addition, they are increasingly used by commercial and industrial companies, on campuses (schools or universities), in hospitals, in military and government installations, and in residential communities.

In interconnected mode, these microgrids form clusters known as Multi-Microgrid (MMG) systems. At the local level, this configuration reduces the instability of the "isolated" mode of microgrids, and facilitates network expansion. Furthermore, the extension of these clusters over a larger geographical area leads to a better application of the Smart grids concept [15].

However, the massive use of smart grids and microgrids, along with the evolution of information and communication technologies, has been accompanied by an exponential increase in the number of vulnerabilities and cyber threats that compromise the security of reliable energy supply.

3. Sustainable energy security

Energy security refers to the assurance of a continuous, reliable, environmentally friendly and affordable supply of green energy to meet the needs of individuals, communities and industry. It is an essential aspect of sustainable development and economic stability, as access to clean, secure and sufficient energy is fundamental to a variety of sectors, including industry, transport, agriculture and health, according to the International Energy Agency (IEA).

Indeed, the concept of energy security encompasses several dimensions, including:

- **Availability** [16] : ensuring a steady supply of energy resources to meet current and future demands.
- **Accessibility** [16] : Ensure that energy is accessible to all segments of society, especially vulnerable populations.
- **Affordability** [16] : ensuring that energy prices remain affordable for consumers.

- **Reliability** [17] : ensuring a stable and constant energy supply, with minimal disruption or downtime.
- **Sustainability** [18] : Ensuring that energy production and consumption practices do not compromise the environment or deplete finite resources.
- **Resilience** [17] : Improving the ability of energy systems to withstand and recover from external shocks, such as natural disasters, geopolitical tensions or **cyberattacks**.

As a result, energy security is a complex and multifaceted challenge [11], influenced by a variety of factors, including geopolitics [19], technological advances, environmental concerns [20] and economic considerations.

4. Cyberthreats and smart grids IT security challenges

4.1. Historical smart grids cyberattacks

Since 1982, several major cyberattacks have affected parts of the global energy system. Electric power systems were among the preferred targets of attackers and terrorists alike:

- **The first major computer attack** on an energy system took place in 1982 in Siberia, and led to the explosion of a gas pipeline.
- **The Blackout of 2003**, which affected the northern part of the American continent (eight American states and southern Canada) due to a malfunction in the general electricity network, had serious economic, financial and health consequences, paralyzing industry, crippling hospitals, and disrupting the operation of airports and public transportation.
- **The 2010 cyberattack**, operated by the "Stuxnet virus", caused major damage to an uranium enrichment plant in Iran. This attack revealed not only unknown vulnerabilities in the energy industry, but also, and above all, the new geo-political-security dimension of this type of attacks.
- **The cyberattack of 2014**, operated by a virus called "Energetic Bear" (a variant of the Stuxnet virus), which destabilized 250 American and European energy companies by successive contagion. This major attack started by infecting just 3 industrial control sites, before spreading.
- **The 2015 attacks** that caused the disconnection of part of the Ukrainian power grid, followed by operation in degraded mode for several weeks. These attacks were carried out using malware known as "BlackEnergy" and "Industroyer", which affected power grid control systems.
- **The Taiwan power grid attack (2020)**: Taiwan's state-owned electricity company "Taiwan Power Co" was the victim of a cyber-attack using ransomware. The attack disrupted some of the company's operations and demanded a ransom to restore systems.

Protecting power grids against cyber-attacks has thus become paramount to any country's security and stability. Governments, energy companies and regulators are increasingly aware of the importance of strengthening the cybersecurity of power infrastructures in order to minimize the risks associated with cyber threats.

4.2. The Cybersecurity of smart grid information systems

Cybersecurity of smart grids involves defining the potential cyber-risks that threaten their information systems, identifying vulnerabilities, and reducing the menace of possible attacks.

In fact, in the field of information systems security, a security risk is defined as the probability that one or more threats (internal or external), would exploit one or more vulnerabilities present in a system (ISO/IEC 27000: 2018 Standard). As for the term "attack", it is simply a threat that takes direct action.

First, the term "vulnerability" is generally attributed to a weakness in an information system (the example of a Backdoor in the information system of an intelligent network). This weakness, if discovered and exploited by an adversary (competitor, malicious person, hacker, etc.), can cause losses and damages to the system.

Regarding threats, they are defined as internal and external factors and circumstances that have the potential to cause damage to the system, by exploiting one or more existing vulnerabilities in its design. A threat would not only be caused intentionally, but also accidentally. For the security of an information system, even errors or hazardous vulnerabilities can degrade performance and paralyze the entire system.

Therefore, to ensure the security of Smart grid data and communications, it is essential to first identify the information system vulnerabilities, before detecting and mitigating the threats.

5. Taxonomy of cyberattacks on smart grid information systems

While the advantages of smart grids over traditional grids are undeniable, smart grids face major challenges in terms of cybersecurity.

Firstly, not only smart grids are vulnerable to cyber-attacks, but so are traditional grids [21]. As an example, a study by Dabrowski demonstrated how malware controlled by a network of zombies could destabilize a traditional so-called "non-smart" power grid.

Secondly, the more complex the network becomes, the greater is its susceptibility to error. For example, if there are more objects connected to the smart grid, there will be more cases of erroneous data being transmitted by faulty sensors [22].

In addition, other sources of smart grid vulnerabilities exist, such as the increasing number of smart devices connected (IoT) to a network, the installation of unverified third-party components, unqualified operating personnel, and the use of unsecured communication protocols.

Moreover, the proper functioning of a smart grid (SG) relies mainly on the two-way communication between power generators and loads on one hand, and between the grid and its sensors installed at various locations on the other. These sensors continuously transmit relevant data to the grid in the form of data packets (information on generation, consumption, voltage and frequency, as well as other energy-related data). This mode of operation makes it vulnerable to numerous cyber-attacks[23].

Generally, cyberattacks on smart grids can be classified as passive or active. Passive attacks are those where the data is not modified, the attacker simply monitors it, while active attacks are more dangerous because the attacker modifies the data or prevents the recipient from receiving it.

Passive attacks are classified into two sub-categories: eavesdropping attacks and traffic analysis attacks. While, active attacks include masquerade attacks, replay attacks, false data injection attacks and denial-of-service attacks [10]. We present here a brief description of some of the major attack classifications affecting Smart grids:

- **Eavesdropping attack:** The attacker can see the data packets exchanged, without altering them.
- **False data injection (FDI) attack:** When the data is modified, but both sender and recipient are unaware of the attacker's modification.
- **Denial-of-service (DoS) attack:** The attacker generates a large number of irrelevant requests to the server, which begins to respond until all its resources are exhausted, at the point where the server is blocked.
- **Man in the Middle (MitM) attack:** the attacker intercepts communication between two nodes without their approval.

In conclusion, power grids, whether traditional or smart, are vulnerable to various types of cyber-attack. It is therefore essential to put in place appropriate security measures and mechanisms to reduce the impact of risks and guarantee the reliability, durability and resilience of these systems.

6. Approaches for detecting and mitigating cyber-attacks on smart grid

Smart grids involve many players, including consumers, electricity companies, grid operators and third-party service providers. Due to the involvement of these multiple stakeholders, smart grid information system management can be a very complicated task, and prone to numerous IT vulnerabilities.

Various approaches have been proposed in the literature to deal with the cybersecurity aspect of smart grids, such as the key management-based approach and the IDS-based approach, detailed in [9] and [24] respectively. However, our research has enabled us to deduce that the most promising yet least documented approaches are those based on machine learning (MLB), cloud computing (CCB), blockchain (BB) as described by Tufail et al [23] ,and finally those based on multi-agent systems [25].

We can briefly summarize the advantages and disadvantages of the different approaches as follows:

First, the machine learning-based approach offers many advantages, such as high accuracy, and continuous, adaptive learning. However, this approach requires large amounts of historical data to train the model, which can be very costly in terms of computation and time.

The other approach is based on cloud computing. This offers a high level of security, low computing costs and low latency. However, it is dependent on bandwidth availability, and is highly reliant on the cloud service provider.

Then, there's the blockchain-based approach, which is considered highly secure, thanks to distributed data storage and the immutability and encryption of all transactions. However, it requires high energy consumption to run all the nodes, and also private blockchains are not yet well secured.

To overcome the limitations of these approaches, an emerging MAS-based cybersecurity approach is rapidly gaining ground, demonstrating significant potential for providing robust cybersecurity solutions for smart grid information systems.

7. Multi-agent systems and smart grids

7.1. Comparison between multi-agent systems and smart grids

From a structural point of view, there are strong similarities between Multi-Agent Systems and a network of interconnected microgrids. This analogy lies in their similar designs and the different types of collaboration they implement.

On the one hand, we've already seen that a smart grid could consist of interconnected microgrids that exchange data and energy as needed. Consequently, it can be deduced that a microgrid can be considered as an essential node within this structure. On the other hand, multi-agent systems are computer systems composed of autonomous entities, called agents that collaborate to solve complex problems.

Moreover, in MAS, each agent is capable of making independent decisions [5] to achieve personal or common goals. Similarly, the Microgrids of a Smart grid can be seen as autonomous nodes [15] capable of generating, distributing and consuming energy independently or collaboratively.

Here is a simplistic representation of this analogy between MASs and Microgrids in structural terms:

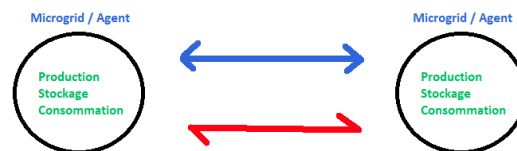


Fig. 1. Diagram showing the analogy between MASs and Microgrids.

Figure 1 illustrates how Microgrids act as autonomous agents/nodes in a decentralized energy environment. Each Microgrid is represented by a circle (Microgrid = agent) with its energy production, storage and consumption capacities/functions. The blue arrow symbolizes the interactions (collaborations and communications) between microgrids, and the red arrow represents the energy flows exchanged on the grid.

From a functional point of view, Microgrids can be assimilated to agents, as both interact and collaborate with each other to achieve clearly defined individual or collective objectives.

On the one hand, multi-agent systems incorporate several types of collaboration. The most important ones were introduced by Jennings and Wooldridge [5], including Coordination, Cooperation, Coalition [6], Competition, Negotiation, Communication, Trust and reputation [26], etc.

On the other hand, collaboration between microgrids is essential to optimize energy management, improve efficiency and guarantee the reliability of a smart grid. Indeed, it allows renewable energy resources to be exploited more effectively, improves resilience and ensures a more reliable and sustainable power supply.

Based on this comparison, we could confirm that there is a strong analogy between multi-agent systems and Smart grids, both structurally and functionally. Indeed, these two concepts share similarities in terms of their architectures and their ability to manage resources and operations in a distributed, intelligent way.

7.2. The use of MASs principals in smart grids

In many studies, it has been shown that the implementation of certain principles specific to multi-agent systems has helped to solve several constraints related to the field of smart grids [1, 2]. For example, the balancing between supply and consumption, the conduct of negotiations between different stakeholders, and the optimization of distributed energy storage [8].

In the field of energy management, an approach based on MAS has been adopted by researchers such as [8], and [3]. For Perera [3] the multi-agent system concept offers a "plug and play" control nature to a microgrid, unlike complex and conventional control techniques. Moreover, according to Nguyen [8], the efficiency of DER management has been improved considerably thanks to the application of the MAS model in the microgrid field.

In the field of information security systems, researches in the area of MAS(s) [7, 9, 10] have been able to address a number of challenges related to the risks of cyber-attacks [25], to the security of communications between agents [27], and the problems of establishing trust and appropriate access controls within the system [28]. However, only a few studies have explored in depth the possibilities offered by the use of multi-agent system security principles for the benefit of smart grid collaboration.

After analyzing and comparing the multi-agent system approach with other cybersecurity strategies for smart grids (in particular, machine learning (MLB), cloud computing (CCB), and blockchain (BB) approaches), we have concluded the preeminence and benefits of this approach.

First, each approach is offering particular strengths suited to specific contexts and constraints. However, thanks to the verified similarities of their architectures, the MAS approach demonstrates strong potential for delivering robust cybersecurity solutions for smart grid information systems.

While there are difficulties in implementing the cybersecurity principles of the MAS approach in Smart Grid information systems, and certain considerations are needed to overcome them (particularly in terms of scalability and inherent limitations). Nevertheless, the practical applications and benefits associated with implementing MAS-based cybersecurity measures in smart grids are greater and more advantageous.

Effectively, MLB offers adaptive security, but requires a lot of historical data and resources. CCB offers high levels of security, but is dependent on cloud service providers and may have bandwidth requirements. BB guarantees security through decentralized ledger technology, but faces power consumption issues and limited search. Whilst, MAS excels in decentralized decision-making and collaboration, enabling it to tackle complex challenges, even though scalability can be little problematic to manage efficiently.

Thus, the MAS-based approach stands out for its adaptability, collaboration and ability to tackle complex problems in smart grids field, offering resilience and flexibility without relying heavily on historical data, bandwidth limitation or additional resources.

8. Conclusion and perspectives

A comprehensive literature review on Multi-Agent Systems and Smart Grids has been conducted. The review focused on smart grids, MASs, and the opportunities of implementing MAS cybersecurity concepts in smart grids systems.

In conclusion, matching the goals and objectives of Smart Grids with the powerful problem-solving capabilities of Multi-Agent Systems (MAS) provides a unique opportunity to strengthen the security of Smart Grid information systems. The integration of multi-agent systems in securing smart grid information systems is considered an innovative and effective approach. Numerous use cases where multi-agent systems can improve smart grid security have been widely discussed by researchers in recent years. The main features of MAS, such as decentralized decision-making processes and collaboration, are recognized for their potential to address the various cybersecurity challenges inherent in smart grids. The integration of MAS security protocols into smart grid information management systems can be used to facilitate secure communication, threat detection and response mechanisms.

In addition, the similarities between these two concepts make it ideal to seamlessly implement MAS-based security. So, it would be useful to further explore the possibilities of using the mechanisms and principles of multi-agent systems, especially to enhance communication security and trust within smart grid information systems.

Furthermore, we have concluded that applying MAS principles in an interconnected microgrid environment would not only secure their communication and collaboration, but it would also optimize the management of information and energy flows.

The purpose of this paper is to attempt to summarize and classify the relevant research on this subject. Thus, in our perspectives, we plan to pursue our research in this direction, exploring in more detail how MAS concepts can be applied to smart grids collaboration and communication security for more robust and resilient energy supply in the face of cyber-attacks.

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